Who made the decision?

From User Clicks to AI Decisions

Full-Stack Observability with React & OpenTelemetry

Glodie Juvenal Pengele /DEV/MTL 2026

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Project X

01



Meet John Doe.

Developer. Ships fast. Uses AI every day.

Project X



BOSS

I have an idea: a booking app. The next best application. Let's build something quick for Friday.



JOHN

JOHN THINKS…

React. Obviously.

I'm a big React fan. React + Vite for the frontend.

+ a small API with Express

Just enough backend to serve the events.

And ship it fast.

My AI agent can write most of it.



lib/events.ts

✦ written by AI agent

```ts
export async function getEvents() {
  const events = await db.query('SELECT * FROM events')

  for (const event of events) {
    event.seatsTaken = await db.query(
      'SELECT count(*) FROM reservations WHERE event_id = $1',
      [event.id],
    )
  }

  return events
}
```

CODE REVIEW

Reviewed by an AI agent

“LGTM. Clean and readable.”

✓ Approved



CI passed

42 tests · 0 failed

LIVE IN PRODUCTION

project-x.example.com/events

Project X EVENTS

React Montréal · Nov 12	Reserve seat
AI Night · Nov 19	Reserve seat
/dev/mtl · Nov 27	Reserve seat



BOSS

Live before Friday? Amazing. Great job, John!

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The bug

02



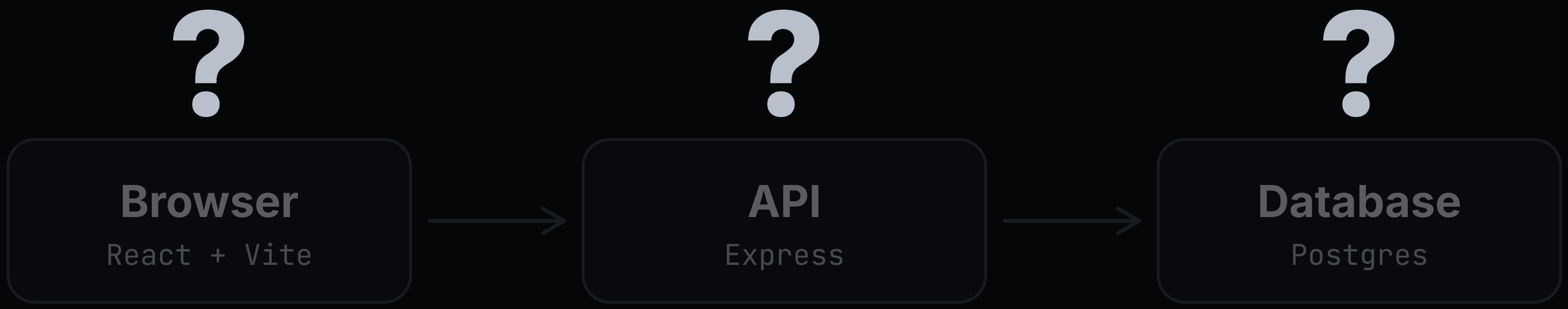
Boss now

Users say the app is slow.



JOHN

The app is slow. But where?



No error. No crash. No idea.



JOHN

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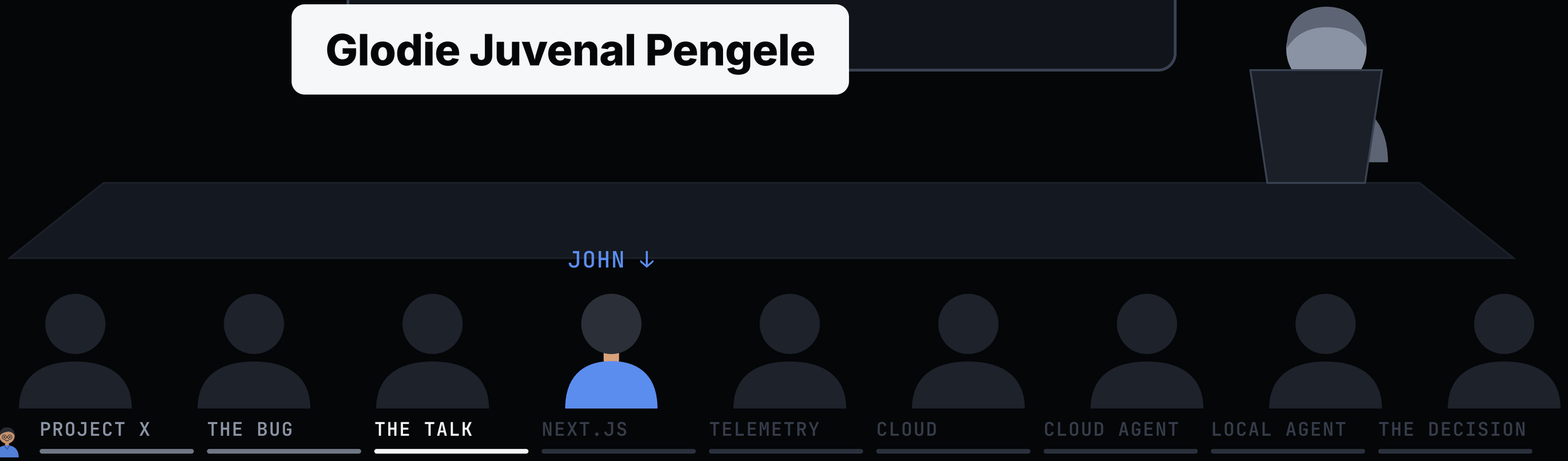
The talk

03

LAST REACT CONFERENCE…

OpenTelemetry in Frontend Applications

Glodie Juvenal Pengele



Observability is your ability to understand what your system is doing from the **outside**, by looking at the data it **emits**.

Not by guessing.
Not by adding `console.logs` and redeploying.

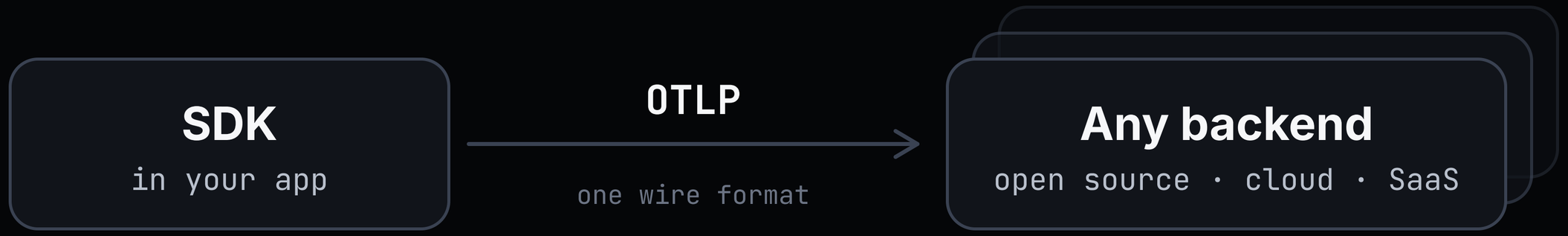


Traces · Metrics · Logs

What happened to this one request, everywhere it went.

OpenTelemetry

An open standard to create, collect, and export telemetry.
Vendor-neutral.



Not for my React app. Not yet...

Browser instrumentation

`@opentelemetry/sdk-trace-web`

EXPERIMENTAL

Server-side tracing

`@opentelemetry/sdk-trace-node`

STABLE



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Next.js

04





BOSS

We need good SEO. People have to find Project X.



JOHN



WHAT JOHN NEEDS

Good SEO

THE BOSS WANTS IT

Keep React

JOHN LOVES IT · V1 IS REACT

Reliable telemetry

FROM THE CONFERENCE

React
stays

+

Next.js
frontend + backend

WHAT JOHN NEEDS

- ✅ **SEO: server rendering**
- ✅ **React stays**
- ✅ **Front + back, one app**
- ✅ **Telemetry on the server**

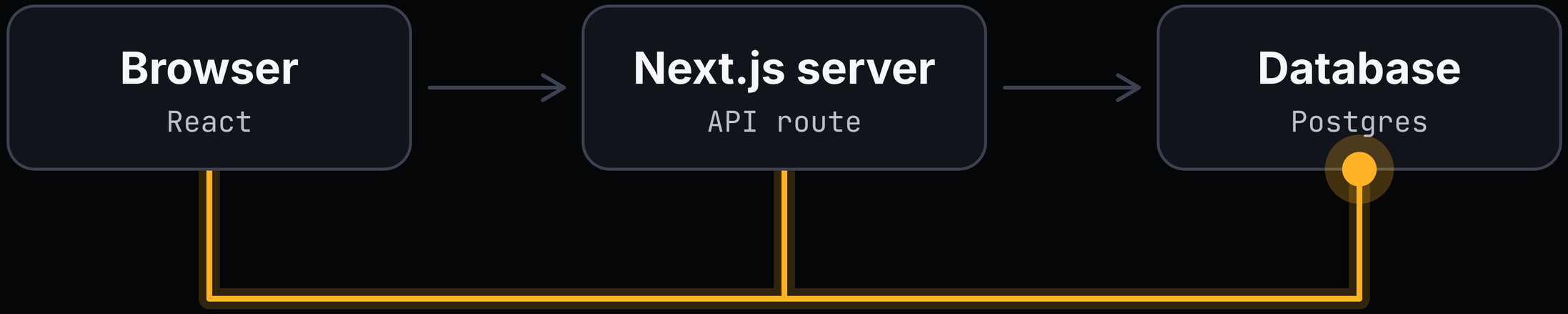
React stays. One app, front and back.



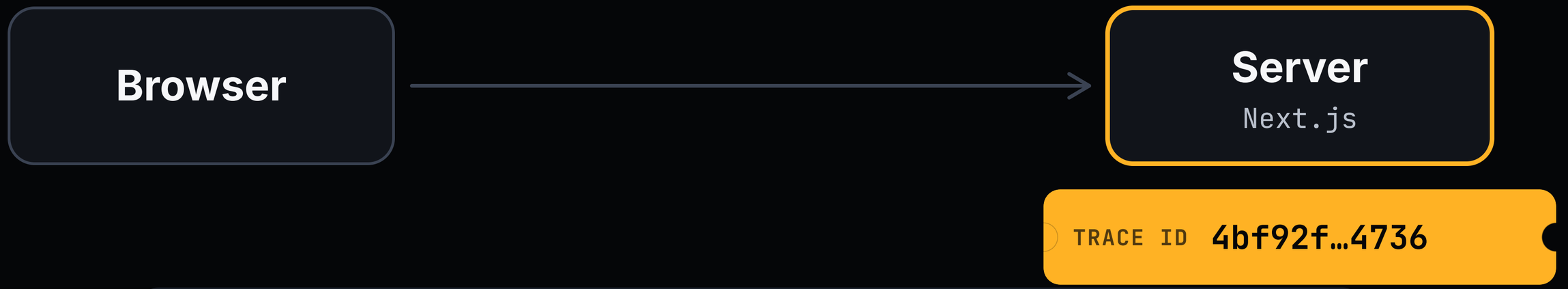
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Telemetry

05



one trace




```
POST /api/reserve HTTP/1.1
host: project-x.example.com
content-type: application/json
traceparent: 00-4bf92f3577b34da6a3ce929d0e0e4736-00f067aa0ba902b7-01
```

Same trace ID on both sides → one trace

instrumentation.ts

```ts
import { NodeSDK } from '@opentelemetry/sdk-node'
import { OTLPTraceExporter } from '@opentelemetry/exporter-trace-otlp-http'
import { getNodeAutoInstrumentations } from '@opentelemetry/auto-instrumentations-node'

export async function register() {
  if (process.env.NEXT_RUNTIME !== 'nodejs') return

  const sdk = new NodeSDK({
    serviceName: 'project-x',
    traceExporter: new OTLPTraceExporter(),
    instrumentations: [getNodeAutoInstrumentations()],
  })

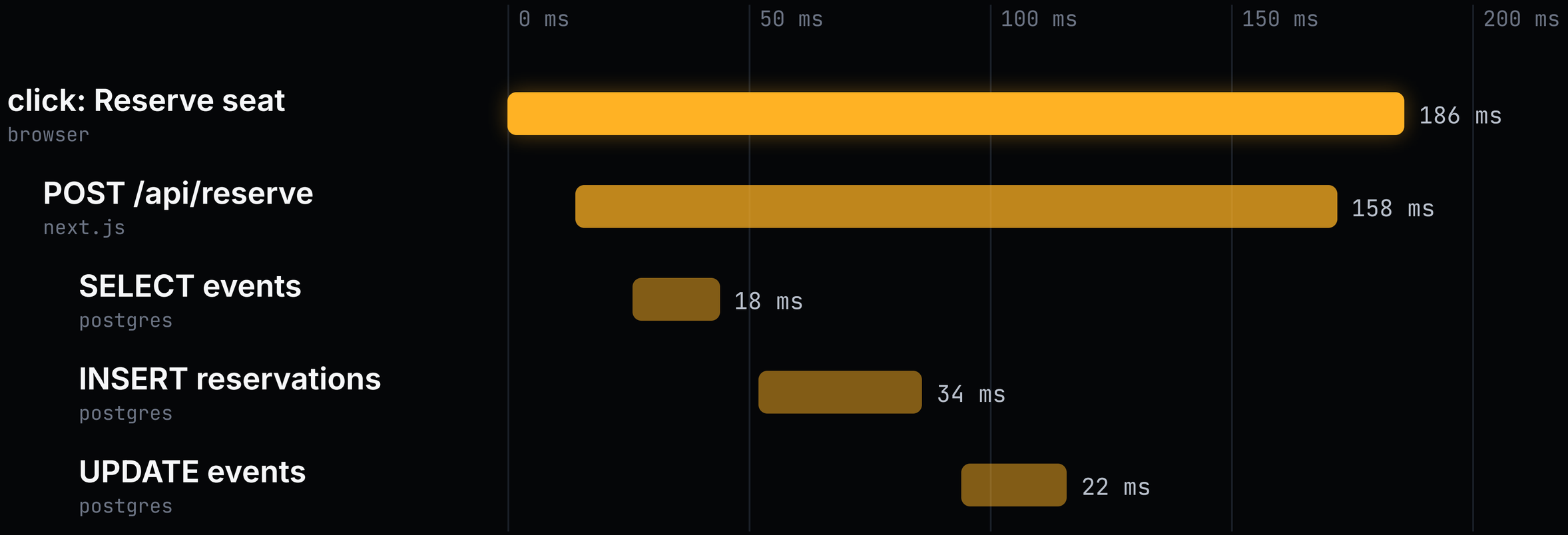
  sdk.start()
}
```



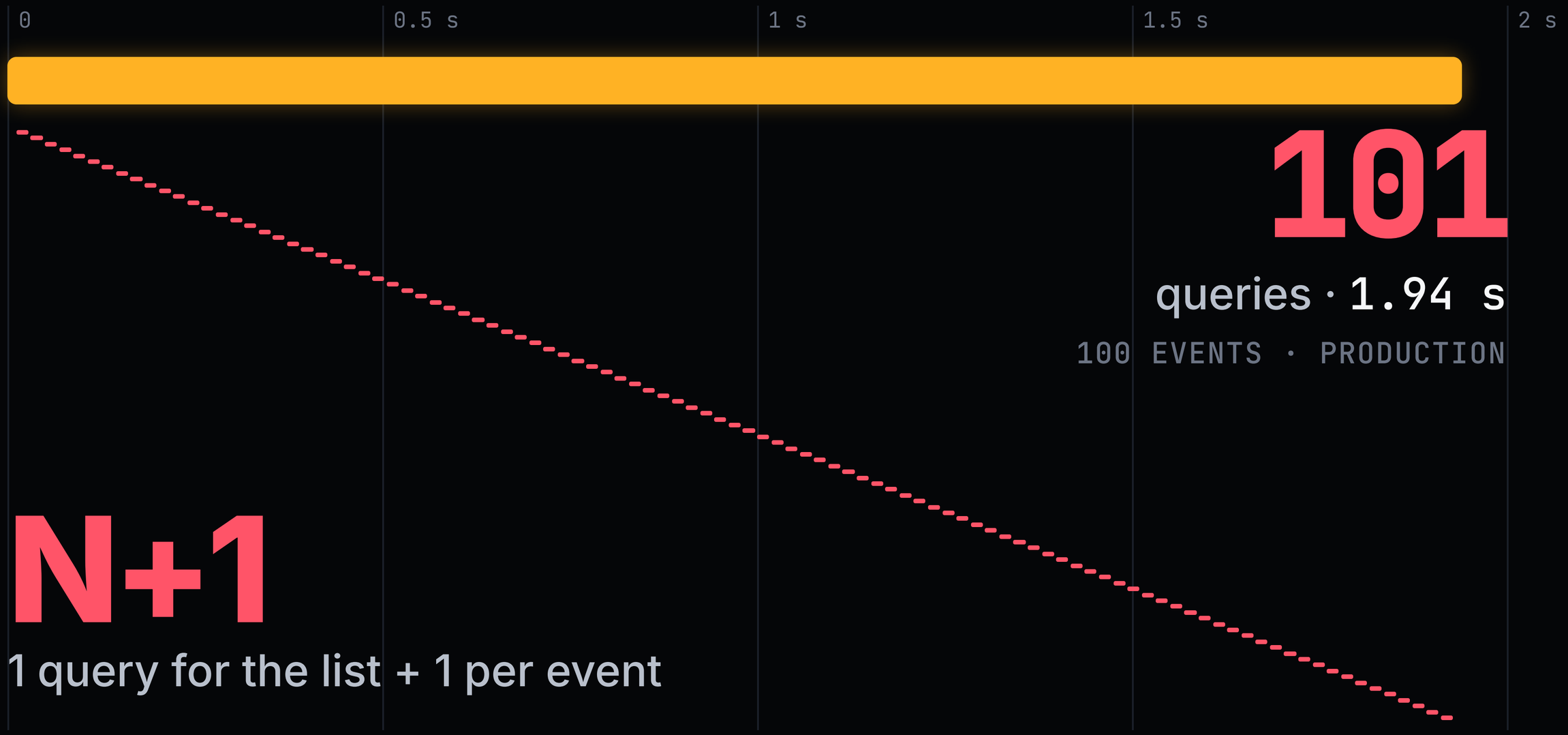
LIVE DEMO

Demo: John's first trace

JOHN'S FIRST TRACE · 4BF92F…4736 · 5 SPANS



GET /events TRACE



N+1

1 query for the list + 1 per event

lib/events.ts

✦ written by AI agent

```ts
export async function getEvents() {
  const events = await db.query('SELECT * FROM events')

  for (const event of events) {
    event.seatsTaken = await db.query(
      'SELECT count(*) FROM reservations WHERE event_id = $1',
      [event.id],
    )
  }

  return events
}
```

Written by the agent. Approved by the agent. Approved by John.



REACT + VITE

? ? ?

no idea

NEXT.JS + OPENTELEMETRY



```
lib/events.ts:5 await db.query(…)
```

the exact line

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Cloud

06





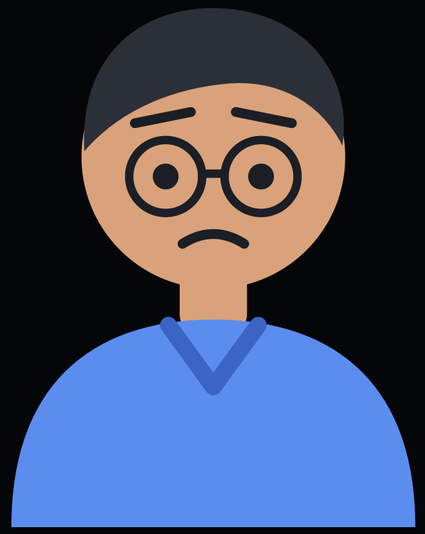
BOSS

We're moving to the cloud.

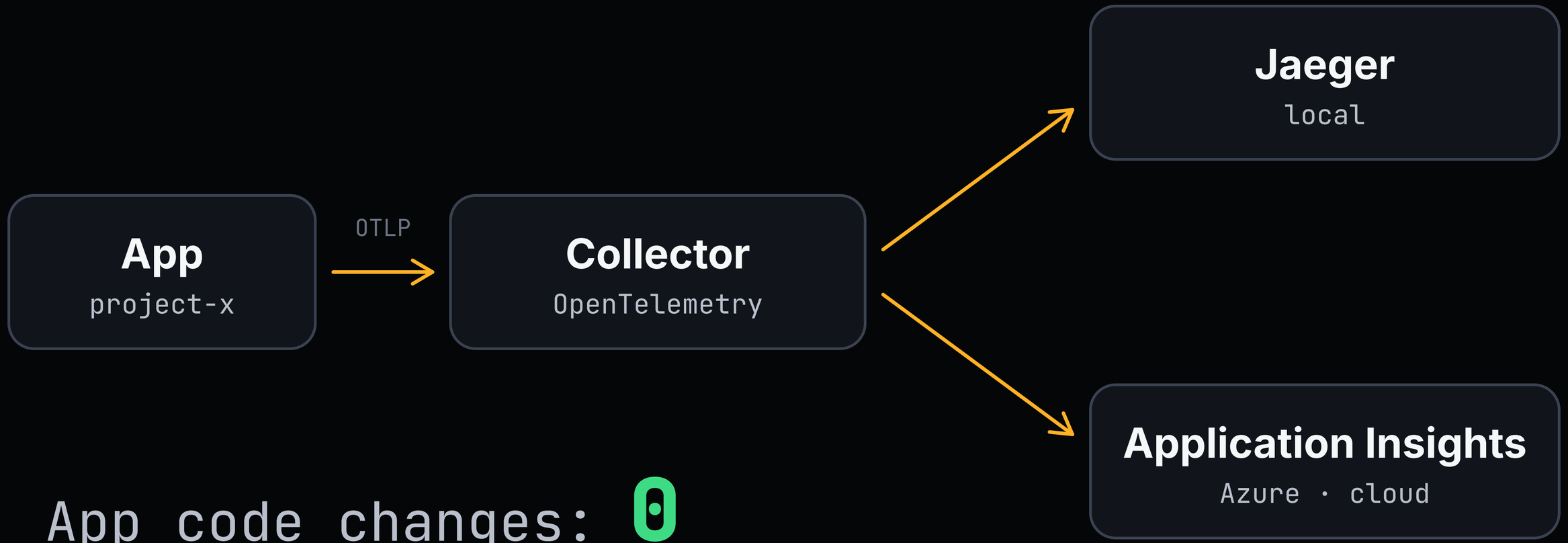


JOHN

Without breaking my workflow?



LOCAL → CLOUD



App code changes: 0

changed: collector.yaml (one exporter)

LIVE DEMO

Demo: same trace, in the cloud

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Cloud agent

07



BOSS

Can AI help us here too?



JOHN

IN THE CLOUD


Traces
Logs
Metrics
Azure SRE Agent
reads your telemetry
FINDING
GET /events: 101 queries per request
SELECT count(*) FROM reservations
WHERE event_id = $1 × 100
N+1 in getEvents() · lib/events.ts

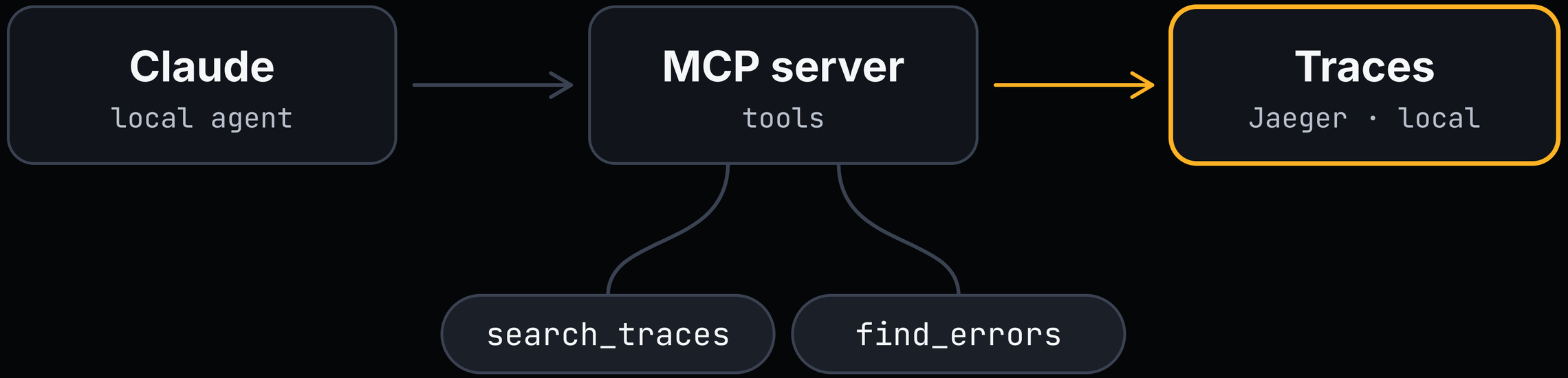
The agent is only as good as your telemetry.

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Local agent

08

ON JOHN'S LAPTOP



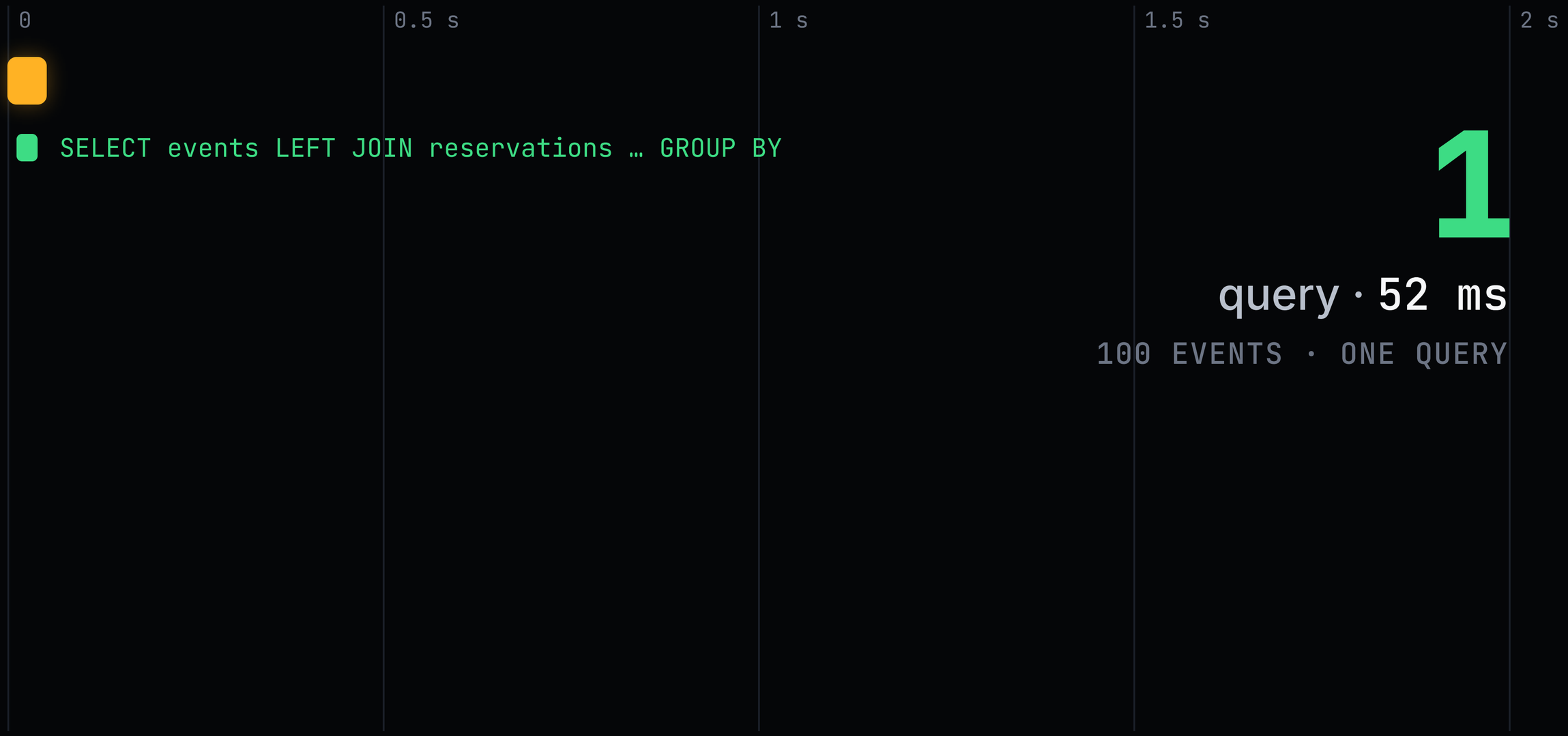
The agent reads the same traces John does.

LIVE DEMO

Demo: Claude debugs locally

```
> Users say the events page is slow. Find out why.
```

GET /events TRACE



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The decision

09

Who made the decision?

AI wrote the code

Observability recorded what it did

Agents read the traces

John decided

TAKEAWAYS

1. Start the trace where the user is.

2. Use OpenTelemetry so your observability outlives your code and your vendors.

3. Give your spans meaning, for humans and for agents.

Observe the AI's decisions.
Don't just trust them.

Thank you

QUESTIONS?

Glodie Juvenal Pengele

your-handle@example.com

github.com/your-handle/project-x

DEMO REPO

public/
feedback-qr.png

FEEDBACK